

Sai Theja Bajjuri

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Summary

Early-career Data Science practitioner with experience building end-to-end ML pipelines, optimizing models, and deploying lightweight inference services. Skilled in Python, R, SQL, and modern ML frameworks, with hands-on work in experiment tracking, model evaluation, and applied statistical modeling for sports analytics and healthcare use cases.

Work Experience

Kanerika Solutions

Dec 2023 - Jul 2024

Data Analyst Intern

Hyderabad

- Processed and validated over 50,000 data records using Python and SQL, improving data formatting accuracy by 28% and reducing analytics reporting time by 17%.

Cigniti Inc

Jul 2023 - Sep 2023

Automation Testing Intern

Hyderabad

- Completed an 80-day internship at Cigniti, where I used Selenium with Java and Git to automate software test cases, increasing test execution speed by 30%.

Education

University at Buffalo - SUNY

Aug 2024 - Jan 2026

M.S., Data Science

- Coursework:** Statistical Methods for DS, Probability, Data Modeling & Query Language, Pattern Recognition

Mahatma Gandhi Institute of Technology, JNTUH

Oct 2020 - Jul 2024

B.Tech, Computer Science & Engineering

Hyderabad

Technical Skills

- Languages:** Python, R, SQL
- ML Frameworks:** scikit-learn, TensorFlow, XGBoost, NumPy, pandas
- Data Tools:** Power BI, Excel, Flask APIs, MLflow Tracking
- Databases:** PostgreSQL, MySQL, SQL-based systems
- Big Data:** Apache Spark (foundational usage)
- MLOps:** MLflow, Git, experiment tracking, basic model packaging & deployment

Projects

Change Point Detection via ChangeForest

- Implemented a sliding-window classifier approach to detect structural changes in multivariate time series.
- Improved detection of non-linear regime shifts by ~12% over classical CPD baselines while keeping inference latency under ~40ms per window.
- Logged ~13,500 windowed samples and model runs in MLflow to compare variants, track metrics, and reproduce experiments.

NBA Player Type Classifier

- Engineered per-game features from 15+ NBA seasons to learn player archetypes based on efficiency, usage, and shooting profiles.
- Built a reusable preprocessing and training pipeline that improved classification performance by ~14% vs. a baseline rules-based approach.
- Deployed a Flask web interface delivering archetype predictions with < 50ms response time for typical user queries.

Healthcare Readmission Risk Predictor

- Modeled 30-day readmission risk using public EHR-style datasets with ~45k patient encounters and structured clinical variables.
- Designed feature sets combining demographics, comorbidities, and visit history, achieving ROC-AUC of 0.87 across validation splits.
- Served the best-performing model via a REST-style Flask API with standardized JSON inputs for integration into clinician-facing prototypes.

Extracurricular

Kaggle

House Prices: Advanced Regression Techniques

- Top 35%: conducted EDA, feature engineering, and model tuning using XGBoost and Random Forest to reduce prediction error vs. baseline models.